Contact

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Top Skills

Embedded Systems

Machine Learning

Computational Electromagnetics (CEM)

Vulcan Flight Technologies

Advancing Aircraft Design Through Open-Engineering and Software Innovation | Aircraft Concept Development | Aerospace Software

New York, United States

Summary

Vulcan Flight Technologies is an independent aerospace engineering volunteered initiative dedicated to democratizing aircraft development through open engineering, advanced simulation, and digital design. We believe the future of aerospace innovation should not be restricted to organizations with multi-billion-dollar budgets. By combining modern engineering tools, open-source development, and systems-level thinking, our mission is to make aircraft concept development more accessible, transparent, and collaborative. Vulcan Flight Technologies specializes in the conceptual design and digital development of next-generation flight vehicles, with a strong focus on Uncrewed Aerial Vehicles (UAVs) tailored to diverse mission requirements, including stealth, speed, endurance, affordability, reconnaissance, loitering operations, and defensive applications.Our goal is to bridge the gap between innovative ideas and engineering execution by providing organizations, engineers, students, and enthusiasts with the tools and resources necessary to explore new aerospace concepts.What We Do- Aircraft Concept Development- Design and develop UAV and aircraft concepts for a variety of mission profiles- Explore trade-offs between performance, survivability, cost, manufacturability, and mission effectiveness- Create scalable digital aircraft architectures- Engineering Support- Assist organizations and teams in designing UAV systems- Develop digital aircraft layouts and systems integrations- Create open-source aerospace hardware and software solutions- Simulation & Concept Testing- Radar cross-section (RCS) analysis- Computational Fluid Dynamics (CFD) analysis- Thermal analysis and heat management studies- Electrical system and propulsion architecture mapping- Performance trade studies and systems-level optimization- Technical Documentation- Every aircraft concept is accompanied by comprehensive technical write-ups that explain:- Mission objectives and design philosophy- Aircraft architecture and operating principles- Performance estimates and engineering trade-offs- CFD and thermal analysis visualizations- Electrical

and propulsion system layouts- Design sketches and supporting diagramsAt Vulcan Flight Technologies, we envision an open aerospace ecosystem where advanced aircraft development is no longer limited by access, cost, or institutional barriers.Open Engineering. Digital Innovation. The Future of Flight.

Experience

Vulcan Flight Technologies

Defense Technologies

January 2024 - Present (2 years 6 months)

About Vulcan Flight Technologies

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